Data Transformation

30-Minute Hands-On Exercise | Week 9 - Section 2C

Learning Objectives: Identify the three tools used for transformation in Power BI | Sequence the transformation pipeline steps in the correct order | Classify data issues as cleaning, filling unknowns, or augmenting problems | Apply the correct transformation technique to real analyst scenarios

Instructions: Complete all four parts in 30 minutes. Refer to your Section 2C notes. Be ready to discuss your answers with the class.

Part 1: Fill in the Blank (~7 minutes)

Complete each sentence with the correct term or phrase from Section 2C.

1.	In Power BI, data transformation is performed using three tools: Cleaning, Filling Unknowns with Lookups, and Augmenting Data.
2.	The first step in the transformation pipeline is to bring raw data into a staging table .
3.	Removing or repairing values like '12 year', '2011', or '999' entered in a years-of-experience field is an example of cleaning column data column data.
4.	Using a zip code to look up the average income or school district for a record is an example of filling unknown with lookups .
5.	Adding a 'Display Name' column by combining First and Last Name, or pulling a Category Name directly into a Products table, are both examples of Augmenting column data column data.
6.	Transforming a continuous birthdate column into labeled age brackets (Child, Teen, Adult, Senior) is called creating a Discretized field.

Part 2: Matching (~6 minutes)

Write the letter of the correct description or example next to each term.

Term / Concept
A. Transformation Pipeline > 5
B. Cleaning Column Data > 6
C. Augmenting Column Data > 3

Description / Example
1. Adding annual rainfall data to city records by joining to an external weather dataset
2. A value that is implausible or inconsistent with the expected format for a column
3. Adding a calculated Order Total field to an Orders table by rolling up line-item amounts

D. Data Abnormality > 2	4. An intermediate table that holds raw imported data before transformation steps are applied
E. Staging Table > 4	5. The ordered sequence of steps used to move data from its source shape to the destination model
F. Filling Unknowns with Lookups > 1	6. Correcting a city name spelling error or removing a record with an impossible age value of 999

Part 3: Put the Steps in Order (~8 minutes)

The eight transformation pipeline steps below are shuffled. Write the correct order number (1-8) in the left column. The first one has been done for you as an example.

Your Order (1-8)	Transformation Step (shuffled)
5	Define column names
2	Remove data that is out of the scope of analysis
1	Bring in raw data to a staging table ← This is Step 1
8	Add row-level calculations (totals, formatted variations, parent/child aggregates)
6	Define column types
3	Remove or repair data that does not conform to column types
4	Fill unknowns / nulls
7	Remove the columns you won't need

Hint: Think about why column names and types are defined after bad data is removed rather than before. Also consider why raw data is staged first before any other action is taken.

Part 4: Scenario Analysis (~9 minutes)

Read each scenario. Identify the transformation technique from Section 2C that applies, and briefly explain why.

Scenario	Transformation Technique + Why
1. A survey dataset has a 'Years Teaching' column with values like 4, 16, '12 year', and 2011. A student wants to calculate average years of experience. What must happen first, and what Section 2C concept applies?	Cleaning Column Data. Before calculating an average, the abnormal values such as "12 year" and "2011" must be repaired or removed. Otherwise the calculation will be inaccurate.

2. A customer table has a 'ZipCode' column, but no income or demographic information. The analyst wants to enrich the dataset so they can segment customers by average household income. What transformation technique applies?	Filling Unknowns with Lookups. The analyst can join another dataset that contains income information by zip code to enrich the existing data.
3. The Products table has a foreign key called 'CategoryID' that points to a separate Category table. A report designer wants to show the actual Category Name directly on product visuals without adding a relationship. What transformation technique solves this?	Augmenting Column Data. Pull the Category Name into the Products table or create a new column so the descriptive information is directly available.
4. A customers table has a 'BirthDate' column. The analyst wants to build a bar chart showing sales broken down by life stage (Child, Teen, Young Adult, Adult, Senior). The raw date cannot be used directly. What must be done?	Create a Discretized field. Convert birth date values into categories like child, teen, young adult, etc.
5. A Gender column contains the value 'Malf' in several rows. A team member wants to automatically correct all such values to 'Male'. What Section 2C concept applies, and what important caution should the analyst keep in mind?	Cleaning Column Data. The value should be corrected to "Male." but before changing data, confirm that "Malf" is actually a typo and not some other valid code. Analysts should avoid making assumptions without verification.

Bonus / Reflection (if time allows)

Section 2C warns: 'Always be very careful about making assumptions with data.' In 3-5 sentences, explain what this means in the context of cleaning data. Give an example of a situation where an 'obvious' fix to bad data could actually introduce a new error or bias into the analysis.